

Spherical Bodies: Shape Evidence

I can use observations from different viewpoints to describe the Sun, Earth and Moon as approximately spherical bodies.

Disc versus sphere

use evidence

More than one viewpoint

use evidence

Approximately spherical

use evidence

[Open the original interactive lesson](#)

What success looks like

- Distinguish a flat disc from a three-dimensional sphere.
- Use more than one viewpoint as shape evidence.
- Explain why approximately spherical is more accurate than perfect sphere.

Retrieve and reconnect

- Last lesson: why did the globe model have a day side and a night side?
- Last week: what stayed fixed while Earth rotated?
- Earlier space learning: name the three bodies in the Sun-Earth-Moon system.
- Lead-in: how could a disc and sphere both look circular from one viewpoint?

A distant body looks like a circle in one photograph. What shape do you predict it is? First ideas are held, not marked.

Think first. Point, say, sign or write your choice. Give one reason.

A. It must be a flat disc.

B. It must be a sphere.

C. One view is not enough; collect shape evidence from other views.

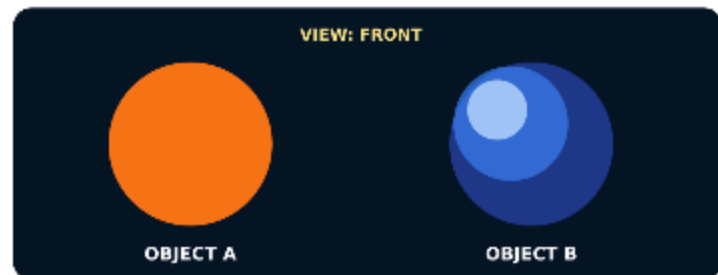
Look closely · what is the evidence?

Observation 1: Object A and Object B both have a round front outline.

Observation 2: after a quarter turn, Object A looks oval and Object B still looks round.

Observation 3: from the edge direction, Object A is a thin line and Object B still looks round.

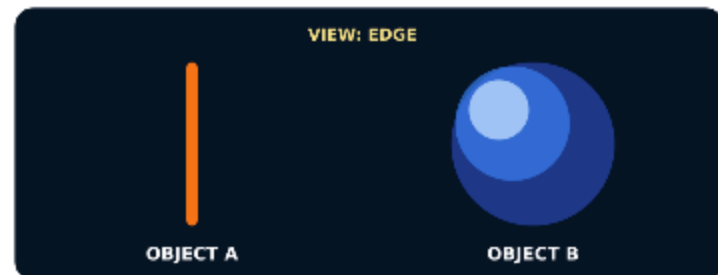
Look closely · what is the evidence?



A disc and a sphere both look round from the front



After a quarter turn the disc looks oval, the sphere still round



From the edge the disc is a line, the sphere still round

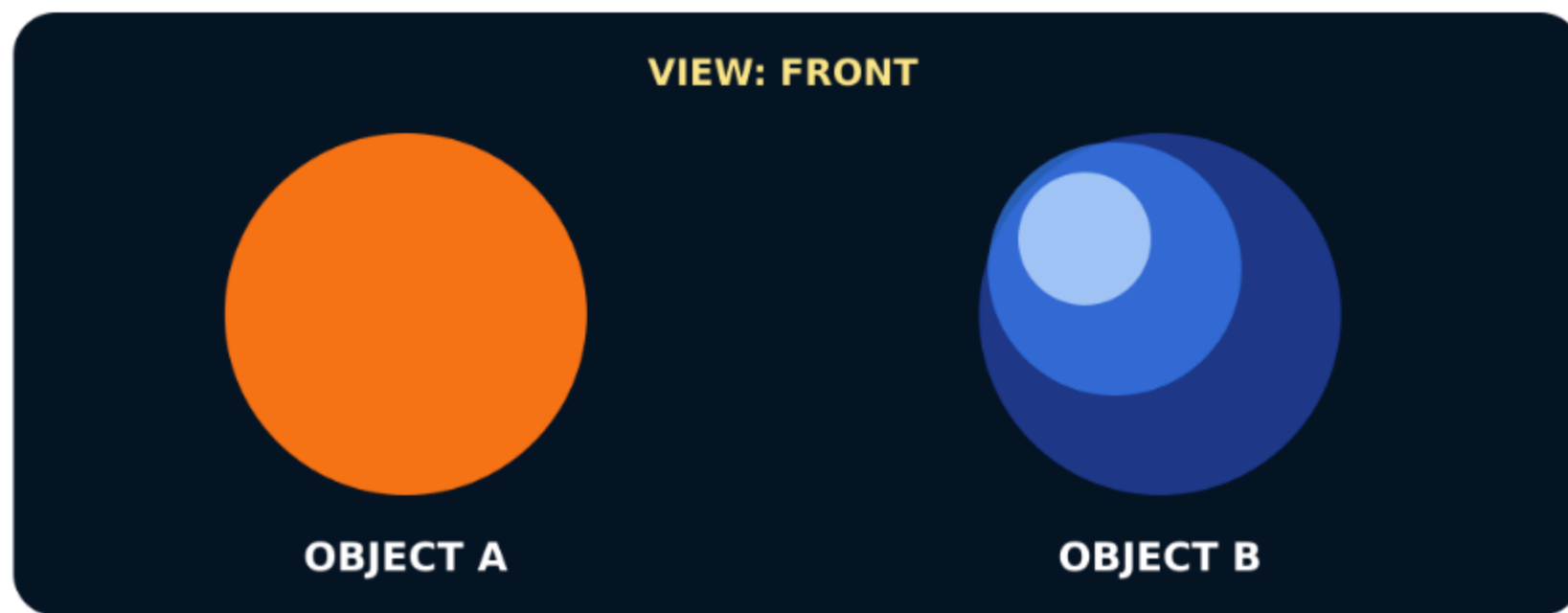
Words we will use

- sphere — ball shape
- disc — flat circle
- approximately — close to

Watch the model

- Hold a flat disc and a ball front-on. Record honestly that both outlines can look circular.
- Turn each object by a quarter turn and then edge-on. Record how the disc outline changes but the ball's outline stays round.
- Conclude from several views: repeated round outlines support a sphere; a single circle does not prove it.

Watch the model



A disc and a sphere seen from the chosen viewpoint

Which observation best separates a sphere from a flat disc?

- A. Both look circular from directly in front.
- B. The sphere keeps a round outline when turned; the disc narrows.
- C. Both can roll in some way.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- The ball-and-disc test shows what two shapes can look like, but it is only a model. Check it against real observations.
- Real Earth and Moon evidence: changing spacecraft views keep showing round whole-body outlines. During a lunar eclipse, Earth's shadow on the Moon is curved.
- Real Sun evidence: its edge is curved and features move across its face as it rotates. Together the observations support approximately spherical bodies; gravity makes large bodies nearly round, rotation slightly flattens them and surface features add small bumps.

Find and repair the misconception

Evidence glitch: Earth looks circular in one photograph, so that photograph proves it is a sphere.

A. Keep it: every circle must be a sphere.

B. Repair it: one view fits a disc too; combine evidence from different viewpoints.

C. Repair it: every distant object is flat.

Shape evidence sorter

Sort model observations and real space evidence. Use both kinds before making the shape claim.

- Select an observation card.
- Commit it to disc, sphere or real space evidence.
- Use the feedback to compare what was seen.
- Freeze model and real evidence, then state the shape claim.

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- Flat disc evidence
- Sphere model evidence
- Real space evidence

Use one model comparison and one real space observation to justify your conclusion.

Your independent mission

Create a multi-view evidence case that distinguishes a disc from a sphere and applies the conclusion to the Sun, Earth and Moon.

Record: A three-view model comparison, one cited real space observation and a conclusion naming the Sun, Earth and Moon as approximately spherical.

Choose your support and challenge

- Supported: Sort six view cards, then complete: A disc changes from ___ to ___. A sphere stays ___.
- Standard: Record three views of each model, then add one real space observation before making the shape claim.
- Stretch: Explain why one circular picture is insufficient and why perfect sphere is also too strong.

Show what you know

Use one model observation and one real space observation to explain why Earth is described as approximately spherical.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.